

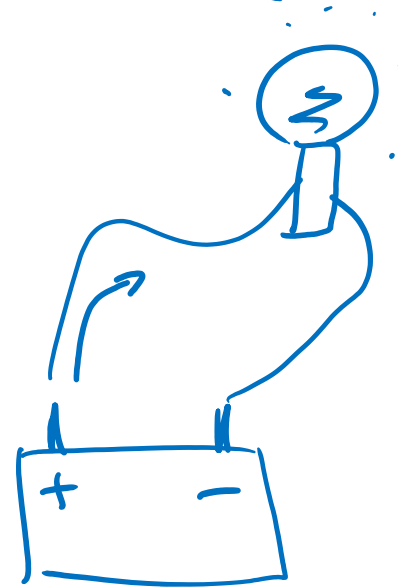
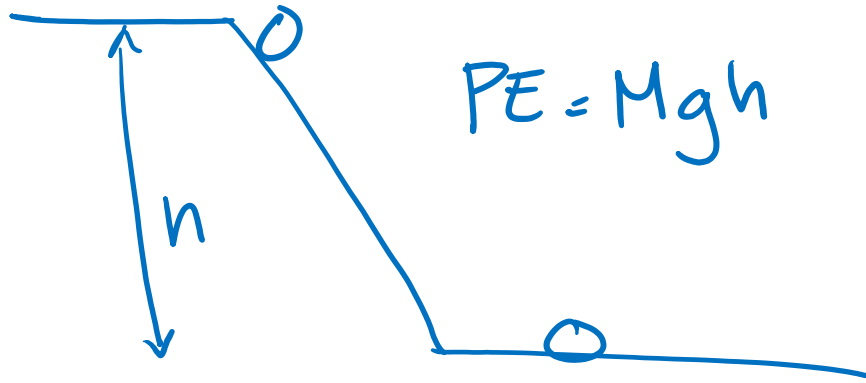


ME 300: Engineering Thermodynamics

Equilibrium

Equilibrium

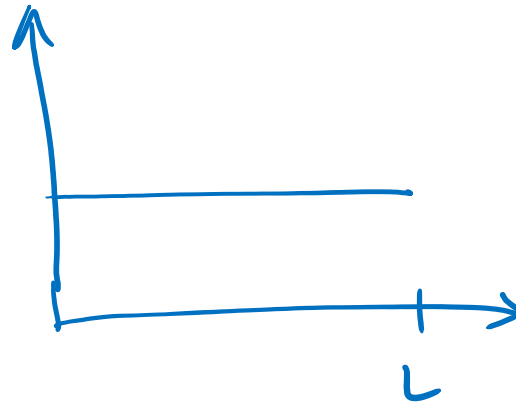
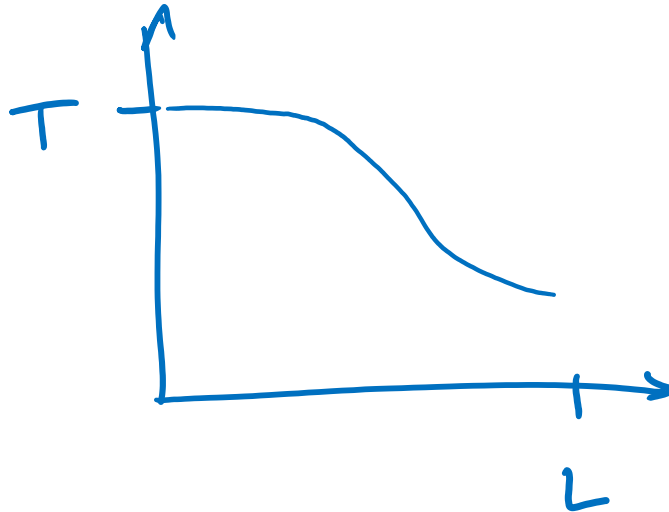
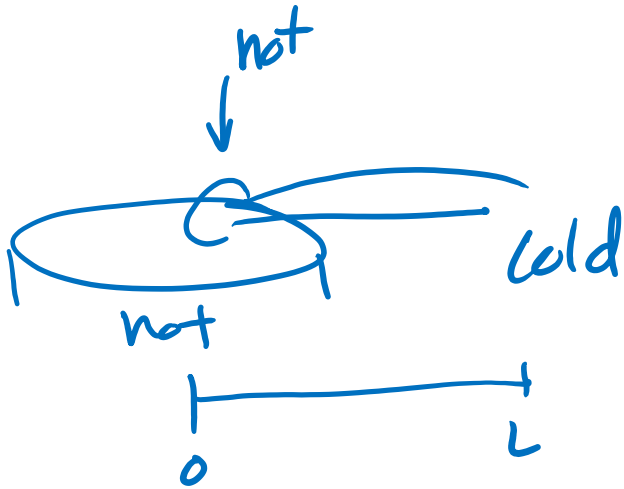
System in equilibrium has no unbalanced potentials.



- thermal
- mechanical
- phase
- chemical

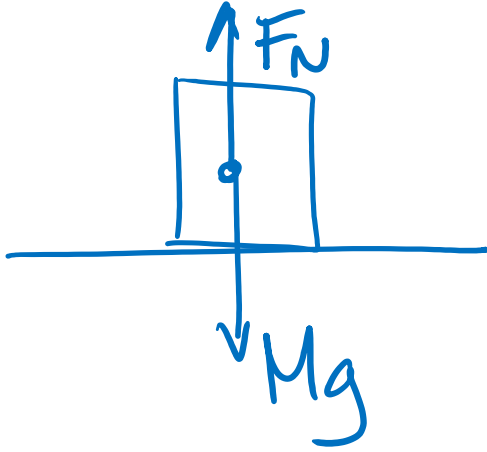
Thermal Equilibrium

System in thermal equilibrium $\rightarrow$ uniform temp
 $\rightarrow$ same temp as its surroundings



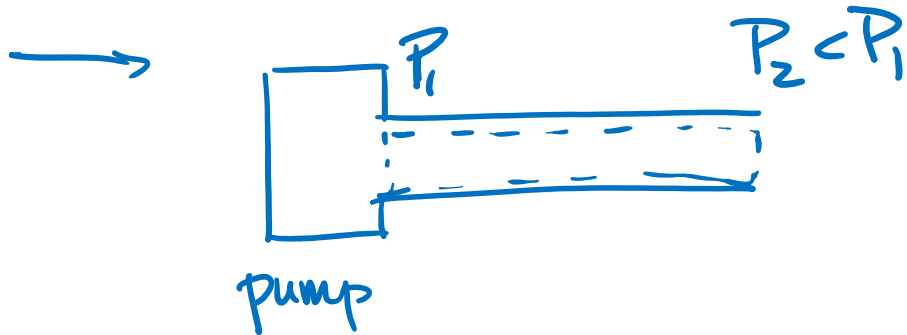
Mechanical Equilibrium

→ no unbalance forces



$$\sum F = F_N - Mg = 0$$

$$\star P = \frac{F}{A}$$



→ turn off pump $P_1 = P_2$

→ cap the pipe



What do I do to
return this system
to equilibrium?

Phase Equilibrium

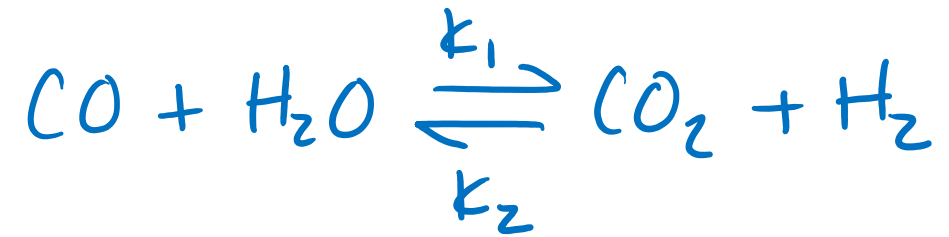
phase equilibrium — amount of each phase is unchanging in time



liquid $\rightarrow$ vapor

Chemical Equilibrium

chem. eq. $\rightarrow$ amount of each species is const.



$k_1 = k_2 \rightarrow$ chem. eq.

Quasi-equilibrium

Quasi-static process is one that happens sufficiently slowly that the time for the system to achieve equilibrium is much shorter than the time scale of the process.

$$t_{eq} \ll t_{process}$$